Weeks 9–10: Group Activity Worksheet

From Problems to AI Agent Workflows

Tutorial Objective: Build on your Assignment 1 analysis. Find **three issues** in your target design. For each issue, propose **3–5 potential solutions**, identify the **technology** each one needs, and write a **step-by-step workflow that an AI agent could follow** to carry out the best solution.

Links to Assessment: This is the core of **Individual Assignment 2 — Strategic Enhancement and Workflow Presentation** (40%, due Week 13).

Two-Week Plan

Week	Activity
9	Briefing: from analysis to strategy; what is an AI agent workflow?
	Part A – Identify and prove three issues
	Part B – 3–5 solutions per issue + technology mapping
	Part C – Prioritise and select one solution per issue
10	Recap: anatomy of a good agent workflow
	Part D – Write three AI agent workflows
	Part E – Test run with an AI tool
	Part F – "Human Agent" peer test
	Part G – Blackboard upload

Materials and Software

☐ Your Assignment 1 report (submitted Week 9) and target design files
☐ Laptop with internet
☐ Access to at least one AI assistant / image tool introduced in class
☐ Contrast checker (e.g. WebAIM Contrast Checker) for accessibility issues
☐ Figma / FigJam / Miro / paper for flowcharts
☐ This worksheet and your Visual Journal

Student Information

Field	Details
English Full Name	
Student ID	
Tutorial Class	
Target Design (from Assignment 1)	
Peer Test Partner	

Part A: Identify Three Issues (Week 9)

A.1 Issue-Finding Lenses

Scan your target design through **every** lens below. Tick the ones where you find a weakness.

Lens	Questions to Ask	☐
Hierarchy & readability	Is the reading order clear? Is anything competing for attention?	☐
Accessibility	Is the colour contrast at least 4.5:1 for body text? Is the text big enough? Can colour-blind users read it?	☐
Consistency	Do spacing, type and colour follow the system everywhere?	☐
Multi-format / responsive	Does it work on mobile, print, social, signage?	☐
Semiotics & culture	Could signs, colours or symbols be misread by some audiences?	☐
Relevance	Does the style feel outdated? Does it still fit the audience?	☐
Production efficiency	Is the design slow or costly to update, translate or scale?	☐
Sustainability & inclusivity	Does it waste materials or exclude any user group?	☐

A.2 Issue Statements

Write each issue as: **"[Who] has trouble [doing what] because [design cause], which leads to [consequence]."**

#	Issue Statement	Lens	Design Principle / Element Affected	Severity *(1–5)*
1				
2				
3				

A.3 Evidence for Each Issue

Each issue needs **proof**, just like in Week 1. Attach an **annotated screenshot** for each issue in your submission.

#	Evidence 1 (observed / measured)	Evidence 2 (comparison, user comment, data, contrast ratio…)	Annotated Image Attached
1			☐
2			☐
3			☐

Optional quick user check: show the design to 2–3 people outside the class for 10 seconds. What did they remember? What confused them?

Part B: Solutions and Technology (Week 9)

B.1 Technology Reference

Category	Examples
Design software	Figma (variables, components, auto layout), Illustrator, Photoshop, InDesign
Generative AI: text	ChatGPT, Claude, Gemini, Copilot (copy, strategy, prompts, research)
Generative AI: image	Adobe Firefly, Midjourney, DALL·E, Stable Diffusion (generative fill, variations)
Generative AI: video / motion	Runway, Pika, Adobe Firefly video
Automation & agents	AI agents with tool access, Figma plugins, Zapier, Make, scripting (Python / JavaScript)
Accessibility & testing	Contrast checkers, colour-blindness simulators, screen readers, user testing
Hardware / output	Digital signage, AR, projection, print finishing, sensors

Name the specific tool AND the specific feature you need (e.g. "Figma variables for colour modes", not just "Figma").

B.2 Solutions for Issue 1

Issue 1: *(copy from A.2)*

#	Potential Solution	Technology Needed (tool + feature)	Effort *(L/M/H)*	Impact *(L/M/H)*	Risks / Limits
1.1					
1.2					
1.3					
1.4					
1.5					

B.3 Solutions for Issue 2

Issue 2: *(copy from A.2)*

#	Potential Solution	Technology Needed (tool + feature)	Effort *(L/M/H)*	Impact *(L/M/H)*	Risks / Limits
2.1					
2.2					
2.3					
2.4					
2.5					

B.4 Solutions for Issue 3

Issue 3: *(copy from A.2)*

#	Potential Solution	Technology Needed (tool + feature)	Effort *(L/M/H)*	Impact *(L/M/H)*	Risks / Limits
3.1					
3.2					
3.3					
3.4					
3.5					

Part C: Prioritise and Select (Week 9)

C.1 Impact / Effort Matrix

Place every solution number (1.1, 1.2, 2.1…) in the matrix.

	Low Effort	High Effort
High Impact	⭐ *Quick wins: do first*	*Big projects: plan carefully*
Low Impact	*Fill-ins: maybe later*	*Avoid*

C.2 Selected Solutions

Issue	Selected Solution #	Why This One (link to the matrix and the evidence)
1		
2		
3		

End of Week 9 progress check (teacher's initials): _____

Part D: Write AI Agent Workflows (Week 10)

D.1 What Makes a Good AI Agent Workflow?

An AI agent follows instructions **literally**. It cannot read your mind. A good workflow is one that **another person or an AI** could follow without asking you any questions.

Component	What to Write
Goal	One sentence: the measurable result
Role	Who the agent acts as (e.g. "senior brand designer")
Inputs	Files, brand rules, data, references the agent receives
Tools	Software / AI tools the agent is allowed to use
Steps	Numbered, one action per step, in order
Decision rules	IF … THEN … rules for choices and problems
Quality checks	Measurable acceptance criteria
Human checkpoints	Where a human must review or approve
Outputs	Exact files, formats, sizes, names
Feedback loop	How results are evaluated and improved
Constraints & ethics	Copyright, brand safety, privacy, disclosure

D.2 Example (for reference only)

Issue: Body text on the poster has 2.8:1 contrast, so it is hard to read. **Solution:** Generate accessible colour variations within the brand palette.

Component	Example
Goal	Produce 3 poster colour variants where all text reaches at least 4.5:1 contrast while keeping brand recognition.
Inputs	Original poster (PNG + Figma file), brand palette HEX list, logo rules
Tools	Figma (variables), contrast checker, an AI chat assistant for palette proposals
Steps	1. Extract all text/background colour pairs. 2. Measure the contrast of each pair. 3. List the pairs below 4.5:1. 4. Propose 3 palette adjustments using brand colours only. 5. Apply each palette as a Figma variable mode. 6. Re-measure all pairs. 7. Export the variants.
Decision rules	IF a pair still fails → darken the text by 10% steps (max 3 times) → IF it still fails, flag for a human. IF a colour leaves the brand palette → reject it.
Quality checks	All pairs ≥ 4.5:1; logo untouched; layout unchanged
Human checkpoint	Designer approves the 3 variants before export
Outputs	`poster_v1/v2/v3.png` (1080×1350) + contrast report table

D.3 Agent Workflow — Issue 1

Selected solution: _____

Component	Details
Goal *(measurable)*	
Agent role	
Inputs	
Tools allowed	
Constraints & ethics	

Steps

Step	Action (one per step)	Tool	Output of Step	Decision Rule *(IF … THEN …)*
1				
2				
3				
4				
5				
6				
7				

D.3 Agent Workflow — Issue 1 *(continued)*

Component	Details
Quality checks / acceptance criteria	
Human checkpoint(s) *(at which step, what is approved)*	
Final outputs *(files, formats, sizes, names)*	
Feedback loop *(how to evaluate and improve)*	
Failure handling *(what if the agent gets stuck?)*	

Agent Prompt

Write the full instruction you would give the AI agent (combine goal, role, inputs, steps, rules and outputs).

D.4 Agent Workflow — Issue 2

Selected solution: _____

Component	Details
Goal *(measurable)*	
Agent role	
Inputs	
Tools allowed	
Constraints & ethics	

Steps

Step	Action (one per step)	Tool	Output of Step	Decision Rule *(IF … THEN …)*
1				
2				
3				
4				
5				
6				
7				

D.4 Agent Workflow — Issue 2 *(continued)*

Component	Details
Quality checks / acceptance criteria	
Human checkpoint(s)	
Final outputs	
Feedback loop	
Failure handling	

Agent Prompt

D.5 Agent Workflow — Issue 3

Selected solution: _____

Component	Details
Goal *(measurable)*	
Agent role	
Inputs	
Tools allowed	
Constraints & ethics	

Steps

Step	Action (one per step)	Tool	Output of Step	Decision Rule *(IF … THEN …)*
1				
2				
3				
4				
5				
6				
7				

D.5 Agent Workflow — Issue 3 *(continued)*

Component	Details
Quality checks / acceptance criteria	
Human checkpoint(s)	
Final outputs	
Feedback loop	
Failure handling	

Agent Prompt

D.6 Master Workflow Diagram

Combine all three workflows into **one system diagram** using the structure:

Input → Process → Output → Feedback

Show where the three workflows share inputs (e.g. brand guidelines), where human checkpoints happen (◆ diamonds), and where results loop back for improvement.

Part E: Test Run with an AI Tool (Week 10)

Run **at least one** workflow (or part of it) with a real AI tool and record what happened.

Item	Details
Workflow tested *(Issue #)*	
AI tool used	
Prompt version 1 *(summary)*	
Result 1 *(what worked / what failed)*	
Change made to the prompt or workflow	
Result 2	
Did the output pass your quality checks?	☐ Yes ☐ Partly ☐ No

What did the test reveal about missing steps or unclear rules in your workflow?

Save screenshots of prompts and outputs for your submission and Visual Journal. Declare all AI use.

Part F: "Human Agent" Peer Test (Week 10)

Swap **one** workflow with your partner. Your partner plays the AI agent: they read **only** your written workflow and try to explain exactly what they would do at each step, **without asking you anything**.

Check (filled in by partner)	Yes / No	Where it broke / comment
The goal is clear and measurable		
Inputs and tools are fully listed		
Every step is a single, clear action		
Decision rules cover likely problems		
Quality checks can be measured		
I know exactly what files to deliver		

Partner name: _____ Number of questions they needed to ask: _____

Fixes I will make:

Reflection

Which issue was the hardest to turn into a clear AI workflow, and why?

Where must a human designer stay in control, and what should never be left to an AI agent?

How will these workflows become your Assignment 2 presentation?

Part G: Blackboard Submission

Upload Checklist

☐ Three issue statements with evidence and **annotated images** (Part A)
☐ Solution tables, 3–5 per issue, with technology (Part B)
☐ Impact / effort matrix and selected solutions (Part C)
☐ **Three complete AI agent workflows** with prompts (Part D)
☐ Master workflow diagram (D.6)
☐ AI test-run log with screenshots (Part E) and peer test (Part F)

File Requirements

Item	Requirement
Format	One combined PDF
File Name	`W09-10_[EnglishFullName]_[StudentNumber].pdf`
Location	Blackboard → SSHD2382 → Weeks 9–10 → AI Agent Workflow
Deadline	Before the start of the Week 11 tutorial

Additional Notes

References & Resources

Record all sources, tools and AI prompts used (APA 7th style). Extensive AI-generated content without citation is plagiarism: